STUDY GUIDE – MIDTERM 2

Exam 2 will cover the following activities and learning goals:

ACTIVITY 9 - SLEEPLESS NIGHTS

- Identify the population and the sample in a statistical study.
- Identify whether a sample is representative of the population.
- Apply the correct symbols/words and distinguish between statistic and parameter when referring to a quantitative variable.
- State the null and the alternative hypotheses in words and in terms of the symbol μ, the long-run mean.
- Predict the center, variation, and shape of the null distribution (distribution of a sample means) from repeated random samples of the same size.
- Use a simulation-based analysis to conduct a test involving the mean of a single quantitative variable.
- State a conclusion about the alternative hypothesis (and null hypothesis) based on the p-value.

ACTIVITY 10 - SLEEPLESS NIGHTS – CONT.

- Calculate the standardized statistic under a theory-based approach
- Use a theory-based analysis to conduct a single test involving the mean of a single quantitative variable.
- Identify whether the validity conditions for a one-sample *t*-test are met.
- Evaluate the strength of evidence against the null hypothesis based on the p-value and/or standardized statistic (*t*).
- Recognize that even with appropriately gathered and analyzed data, there is a chance of arriving at an incorrect decision (Type I and Type II errors).

ACTIVITY 11 – LEND ME A HAND

- Approximate and interpret a 95% confidence interval for a population proportion by using the 2SD method.
- Compute and interpret a confidence interval for a population proportion using a theory-based approach, including checking validity conditions.
- Infer how each part of the confidence interval changes when the confidence level or sample size change.

ACTIVITY 12A - DRINKING CALORIES

- Approximate and interpret a 95% confidence interval for a population mean by using the 2SD method.

Modified from *Introduction to Statistical Investigations*, N. Tintle, B. Chance, G. Cobb, A. Rossman, S. Roy, T. Swanson, J. VanderStoep (John Wiley & Sons, 2020)

- Compute and interpret a confidence interval for a population mean using a theory-based approach, including checking validity conditions.
- Infer how each part of the confidence interval changes when the confidence level or sample size change.

ACTIVITY 12B - HAVE A NICE TRIP

- Identify which variable is the explanatory variable and which is response in a study involving two variables.
- Identify potential confounding variables and explain how they provide an alternative explanation for the observed association between the explanatory variable and the response variable.
- Identify whether confounding variables are balanced out between groups.
- Explain that random assignment gives us the ability to draw cause–effect conclusions because it ensures that treatment groups have similar characteristics.
- Identify whether a study uses random assignment and/or random sampling and the implications of these design decisions on the conclusions that can be drawn.

ACTIVITY 13 - MURDEROUS NURSE?

- Organize counts into a two-way table, when data are available on two categorical variables for the same set of observational units.
- Calculate conditional proportion of successes, for different categories of the explanatory variable
- Use conditional proportions to decide whether there is preliminary evidence of an association between the explanatory and response variables.
- Create a segmented bar chart and mosaic plots to display data available on two categorical variables for the same set of observational units.
- Calculate and interpret the difference in conditional proportions.
- Calculate and interpret the relative risk.
- Calculate and interpret the odds-ratio.

ACTIVITY 14 - TAGGING PENGUINS

- State the null and the alternative hypotheses in terms of “no association” versus “there is an association” as well as in terms of comparing population proportion of success for two categories of the explanatory variable (i.e., π_1 and π_2) when exploring the relationship between two categorical variables.
- Implement the 3S strategy: find a statistic, simulate, and compute the strength of evidence against observed study results happening by chance alone.
- Describe how to use cards to simulate what outcomes (in terms of difference in conditional proportions and/or relative risk) are to be expected in repeated random assignments, if there is no association between the two variables.

- Conduct a simulation-based analysis when evaluating claims about two population proportions.
- Find and interpret the p-value for a test of two proportions.
- State a complete conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the standardized statistic and/or p-value.
- Identify whether the validity conditions for a two-sample z-test are met.
- Conduct a theory-based approach when evaluating claims about two population proportions.
- Use the Theory-based method to find a 95% confidence interval for the difference in population proportions of success for two groups, and interpret the interval in the context of the study.

ACTIVITY 15 - LINGERING EFFECTS OF SLEEP DEPRIVATION

- State the null and the alternative hypotheses in terms of "no association" versus "there is an association" as well as in terms of comparing long-run means for two categories of the explanatory variable (i.e., μ_1 and μ_2).
- When comparing two quantitative distributions, identify which has the larger mean, median, and standard deviation.
- Identify whether there is likely an association between a binary categorical variable and a quantitative response variable.
- Implement the 3S strategy: find a statistic, simulate, and compute the strength of evidence against observed study results happening by chance alone.
- Describe how to use cards to simulate what outcomes (in terms of difference in sample means) are to be expected in repeated random assignments, if there is no association between the two variables.
- Conduct a simulation-based analysis when evaluating claims about two long-run means.
- Find, interpret, and/or evaluate the p-value for a test of two long-run means.
- Use the 2SD method to find and interpret a 95% confidence interval for the difference in long-run means.
- State a complete conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the p-value and the study design.

ACTIVITY 16 – PLATE UP!

- State the null and the alternative hypotheses in terms of comparing population means for two categories of the explanatory variable (i.e., μ_1 and μ_2).
- Implement the 3S strategy: find a statistic, simulate, and compute the strength of evidence against observed study results happening by chance alone.
- When comparing two quantitative distributions, identify which has the larger mean, median, and standard deviation.
- Conduct a simulation-based analysis when evaluating claims about two population means.
- Find, interpret, and/or evaluate the standardized statistic and the p-value for a test of two population means
- Conduct a theory-based analysis when evaluating claims about two population means.
- Identify whether the validity conditions for a two-sample *t*-test are met.

- Use the theory-based method to find and interpret a 95% confidence interval for the difference in population means.
- State a complete conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the standardized statistic and/or p-value and the study design.

Below is the specific information that you need to know about each chapter of the textbook:

FROM CHAPTER 2 YOU SHOULD BE ABLE TO

- Identify the population of interest in a research study
- Identify and distinguish between the population (entire group of observational units of interest) vs. sample (those observational units from the population we record information for)
    - Parameter as a numerical summary of a population
    - Symbols for statistics vs. parameters
- Identify the cause and direction of bias in a biased sampling method
    - How can you tell whether a sampling method is biased from a distribution of sample statistics?
- Be able to identify/describe a sampling frame
- Discuss how to use a sampling frame to select a simple random sample (SRS)
    - Specify a sampling frame in a particular context (i.e., the physical list of members of population like the phone book)
    - Number the units in the frame, use a random number mechanism to generate ID numbers, report the members of the sample
- Explain the purpose of random sampling (produce a sample representative of population)
    - Long-run mean of sample statistics will be equal to the population parameter
    - Allows you to generalize conclusions (significant or not!) from the sample to the population.
- Discuss how the amount of "random sampling error" depends on the sample size
    - Larger samples are more precise = statistics cluster more closely around the parameter
      Note that this does **not** depend on the population size as long as the population size is much, much larger than the sample size (10x)
- Interpret a histogram or dotplot/ compare distributions of quantitative data
    - Eyeball the center and spread of the distribution, identify shape as roughly symmetric or not, skewed left vs. skewed right
    - Interpret mean as a measure of center (include measurement units).
        - Use the word "typical" when interpreting the mean or median in context
    - Interpret standard deviation as a measure of spread (include measurement units)
        - Be able to compare (or conjecture) how the spread of two distributions compares from the graphs/nature of variables involved
    - Always put your comments into the context of the study
- Decide which graphs (bar graph vs. dotplot/histogram) to use with which type of variable (categorical vs. quantitative)
- The statistical significance of an observed sample mean based on the null distribution
    - State null and alternative hypotheses about μ

    - Explain what determines the shape, center, and variability of the null distribution of sample means
    - Calculate the standard error of the sample mean $SE(\bar{x}) = \frac{s}{\sqrt{n}}$
        - "Standard error" just means it is an estimate of a theoretical standard deviation. You can pretty much ignore this distinction and just think in terms of standard deviation.
    - Calculate and interpret the standardized statistic (t-statistic)
    - Determine whether a distribution of sample means is expected to be normal or approximately normally distributed
    - If validity conditions are met, can use **Theory-Based Inference** applet to obtain p-value
- Define Type I and Type II errors in context
- Identify when you could commit a Type I error.
- Identify when you could commit a Type II error.

FROM CHAPTER 3 YOU SHOULD BE ABLE TO

- Confidence Intervals: Determine a range of plausible values for the population parameter (π and μ)
- Obtain and interpret value of standard error of distribution of sample proportions
    - Using formula $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
    - Understand that sample size affects variability of sample proportions
- Obtain and interpret value of standard error of distribution of sample means
    - Using formula $\frac{s}{\sqrt{n}}$
    - Understand that sample size affects variability of sample means
- Approximate a 95% confidence interval using the "2SD Method"
    - A 95% confidence for the population proportion (or long-run proportion), can approximate with $\hat{p} \pm 2^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
    - A 95% confidence for the population mean (or long-run mean), can approximate with $\bar{x} \pm 2^* \frac{s}{\sqrt{n}}$
- Be able to interpret a confidence interval referring to the parameter of interest and the population
- Discuss whether/how sample size affects confidence interval (midpoint, width)
- Discuss whether/how sample statistic affects confidence interval (midpoint, width)
- Discuss whether/how confidence level (e.g., 95%) affects confidence interval (midpoint, width)
- Define margin-of-error (half-width of interval, the "plus or minus" part)
- Categorical data: Can use Theory-Based Inference applet to obtain confidence interval based on normal null distribution (one-sample z interval)
    - Given summary data or data file
    - Need large sample size (at least 10 successes and at least 10 failures in the sample) for this approach to be considered valid
- Quantitative data: Can use Theory-Based Inference applet to obtain confidence interval based on normal null distribution (one-sample t interval)
    - Given summary data or data file

    - Need large sample size (at least 20 in the sample) and not strongly skewed sample distribution or normally distributed data.
- Explain factors that influence the width of a confidence interval

FROM CHAPTER 4 YOU SHOULD BE ABLE TO

- Classify the roles of variables as response or explanatory
- Identify a confounding variable and explain how it is confounding
    - Highlight the connections to both the explanatory variable and the response variable
    - Distinguish confounding variables from other variables
- Classify studies as observational or experimental
    - Highlight the imposition of the explanatory variable (or not)
    - Realize all observational studies have the potential for confounding variables
    - A randomized experiment allows one to potentially draw cause-and-effect conclusions (because there should not be any confounding variables)
- Explain the purpose of random assignment (create equivalent treatment groups)
    - Allowed to conclude causation (if results are statistically significant)
- Identify the allowable scope of conclusions based on the study design
    - Cause and effect vs. confounding variables
    - Generalizability vs. sampling bias

FROM CHAPTER 5 YOU SHOULD BE ABLE TO

- Create a two-way table
    - Put the explanatory variable as the columns
- Interpret segmented bar graph and mosaic plots
- Calculate and interpret conditional proportions
    - Be clear on how you are conditioning (probability of what given what?)
- State null and alternative hypotheses for comparing proportions
    - Parameter: Now we have two population proportions! The parameter is now the **difference** in population proportions (or difference in long-run proportions). Don't forget to complete with context and description of the population.
    - In symbols and in words
- Describe how to conduct the simulation by hand.
    - Explain the reasoning behind the card shuffling simulation
- Obtain an approximate p-value from the Two Proportions applet or from a given null distribution.
- Calculate and interpret the p-value.
    - Be careful not to confuse an EVALUATION of the p-value with an INTERPRETATION of the p-value.
- Discuss the effect of sample sizes on the p-value
- Obtain informal confidence interval using SD_{NULL}
- If valid, conduct a two-sample z-test (theory-based method) to compare two proportions.
    - Theory Based Inference applet.
    - Check validity conditions
    - Calculate and interpret the standardized statistic: $z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$

- If valid, calculate a two-sample z-interval for the difference in two proportions.
    - statistic $\pm$ margin-of-error
    - Theory Based Inference applet.
- Approximate a 95% confidence interval using the "2SD Method":

  $(\hat{p}_1 - \hat{p}_2) \pm 2 * \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
- Interpret a confidence interval for the difference in two proportions (difference/direction, population, proportions). Clearly state which population (or long-run) proportion is higher/smaller.
- Be able to discuss factors that affect test statistic, p-value, and confidence interval (midpoint and width)
- Describe the relationship between confidence interval and test of significance.

FROM CHAPTER 6 YOU SHOULD BE ABLE TO

- Compare and contrast two distributions of a quantitative variable
- State null and alternative hypotheses for comparing means
    - Parameter: Now we have two population means! The parameter is now the **difference** in population means (or difference in long-run means). Don't forget to complete with context and description of the population.
    - In symbols and in words
- Describe how to conduct the simulation by hand.
    - Explain the reasoning behind the card shuffling simulation
- Obtain an approximate p-value from the Multiple Means applet or from a given null distribution.
- Calculate and interpret the p-value.
    - Be careful not to confuse an EVALUATION of the p-value with an INTERPRETATION of the p-value.
- Obtain informal confidence interval using SD from simulated null distribution
- If valid, apply a two-sample t-test (theory-based method) to compare two means
    - Theory Based Inference applet.
    - Check validity conditions.
    - Interpretation of standardized statistic: $t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$
- If valid, calculate a two-sample t-interval for the difference in two means.
    - statistic $\pm$ margin-of-error
    - Theory Based Inference applet.
- Approximate a 95% confidence interval using the "2SD Method":

  $(\bar{x}_1 - \bar{x}_2) \pm 2 * \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
- Interpretation of confidence interval for the difference in two population means (difference/direction, population, means). Clearly state which population (or long-run) mean is higher/smaller.
- Be able to discuss factors that affect test statistic, p-value, and confidence interval (midpoint and width).
- Describe the relationship between confidence interval and test of significance.